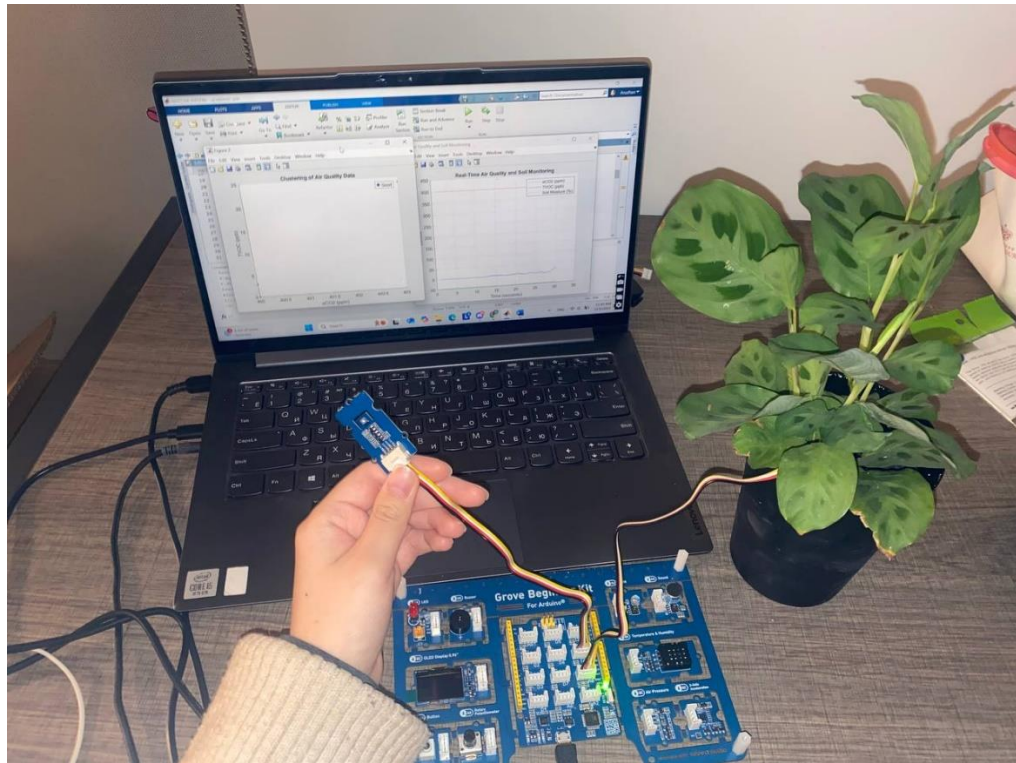


Real-Time Environmental Monitoring Using Sensors and Machine Learning

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Introduction

Environmental challenges like poor air quality and soil degradation significantly affect human health and agricultural productivity. This project integrates real-time monitoring of air quality and soil moisture with predictive modeling of eCO₂ levels, providing an innovative solution for environmental management. By combining sensor data with predictive analytics, the system not only reports current conditions but also forecasts trends, offering actionable insights for proactive decision-making. Predictive analytics has been widely recognized as a key technology in environmental monitoring, particularly for enhancing response capabilities to air quality fluctuations.

Context

This project directly addresses Sustainable Development Goals (SDGs) 13 and 15, which emphasize combating climate change and promoting sustainable use of terrestrial ecosystems, respectively. Accurate monitoring and forecasting of environmental parameters are crucial for achieving these goals. Predictive modeling enhances the ability to anticipate and mitigate risks, such as rising eCO₂ levels, which contribute to poor indoor air quality and broader climate challenges. Research shows that combining real-time sensor data with predictive analytics can improve resource management and environmental policy implementation. The integration of such technologies aligns with the global push for data-driven solutions to pressing ecological issue.

Technical Requirements / Specifications

System Requirements

1. Real-Time Data Acquisition:

- Measure air quality parameters, including equivalent CO2 (eCO2) in ppm and total volatile organic compounds (TVOC) in ppb, using the SGP30 sensor.
- Measure soil moisture levels as a percentage using an analog soil moisture sensor.

2. Data Analysis and Visualization:

- Plot real-time sensor data for air quality and soil moisture on MATLAB using animated graphs.
- Compute and display a correlation matrix to analyze relationships between air quality and soil moisture.

3. Data Processing:

- Normalize soil moisture readings from raw voltage data.
- Apply clustering (k-means) to classify air quality levels (Good, Moderate, Hazardous).
- Perform predictive modeling (linear regression) to predict eCO2 levels based on soil moisture and TVOC.

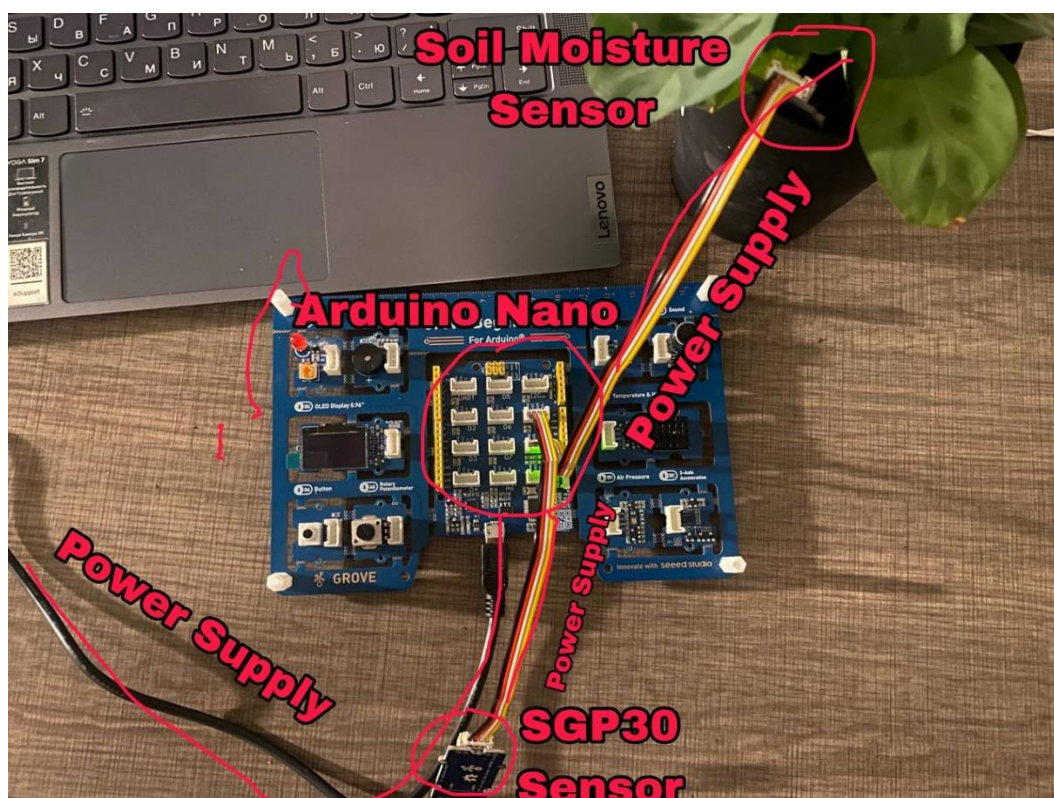
4. Alerts and Anomalies:

- Generate alerts for critical thresholds, such as high eCO2 levels (>1000 ppm) or abnormal soil moisture levels (<20% or >80%).

5. Hardware Integration:

- Use Arduino Nano as the microcontroller for reading sensor data.
- Utilize MATLAB for data processing, visualization, and analysis.

Schematic Diagram



Mathematical Equation

Soil Moisture Normalization

$$\text{Moisture Percent} = 100 - ((\text{Raw Voltage} - V_{\min}) / (V_{\max} - V_{\min}) * 100)$$

$V_{\min}=0$: Minimum sensor voltage(dry soil).

$V_{\max}=3.3$: Maximum sensor voltage(saturated soil).



Procedure

1. Arduino Integration:

- Initialize the Arduino in MATLAB using the arduino and sgp30 objects.
- Write code to read sensor values from the SGP30 and soil moisture sensor.

2. Data Normalization:

- Implement the formula to normalize soil moisture readings from raw voltage values.
- Map normalized values to percentages for easy interpretation.

3. Real-Time Visualization:

- Create animated plots in MATLAB for eCO₂, TVOC, and soil moisture levels.
- Add labels, legends, and dynamic updates to make the visualization user-friendly.

4. Correlation Analysis:

- Store sensor data and timestamps in arrays.
- Calculate the correlation matrix to find relationships between soil moisture, eCO₂, and TVOC, using Matlab Statistics and Machine Learning Library.

5. Clustering:

- Apply the k-means clustering algorithm in MATLAB to classify air quality levels into “Good,” “Moderate,” and “Hazardous” based on eCO₂ and TVOC.
- Visualize clusters using scatter plots.

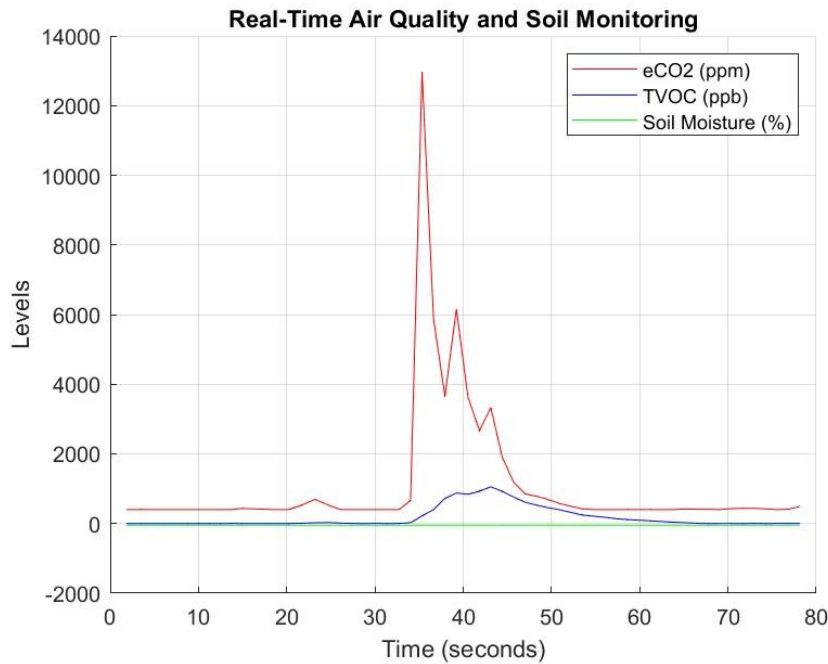
6. Predictive Modeling:

- Fit a linear regression model to predict eCO₂ levels based on soil moisture and TVOC.

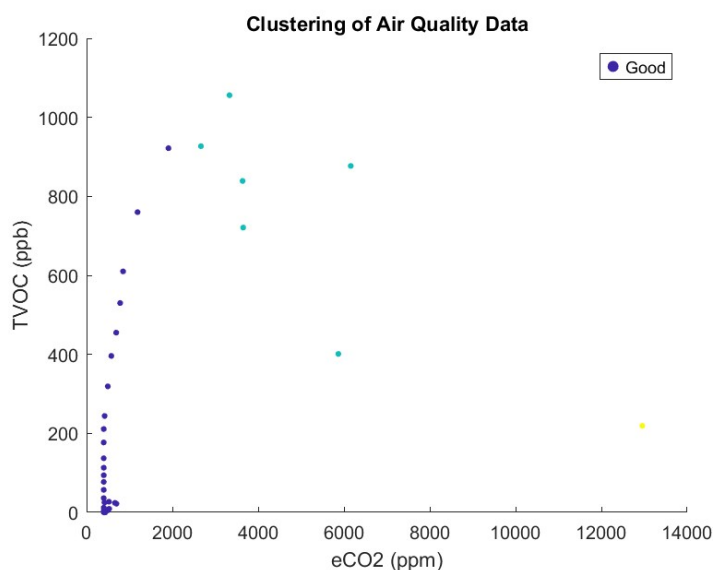
- Display model statistics and use it for predictions with new input data.
- 7. Anomaly Detection and Alerts:**
- Set thresholds for high eCO₂ levels and abnormal soil moisture values.
- Add alerts in the MATLAB command window to notify the user of anomalies.

(See Figure 1. Flow-chart below)

Test



To ensure the system worked correctly, I triggered alerts by manually simulating high eCO₂ levels (>1000 ppm) by exhaling near the sensor and abnormal soil moisture levels (<20%) by holding the sensor in the air. I received following alerts in the Matlab command window: 'Alert: Soil moisture is critically low!', 'Alert: High eCO₂ levels!'. eCO₂ Range showed 400 ppm (ambient level) to 1200 ppm (simulated polluted environment).



Clustering of air quality data showed that initial values are "Good" for health (purple dots), whereas when dots reached the peak, the color of dots changed to light blue, indicating

the hazards for health. However, for the clustering to be more precise, the system needs more time and large data sets.

Predicted eCO2 model also showed moderate correlation. For instance, three consecutive values: 4758.4741, 4830.258, 4909.0021. Last one demonstrated in the picture below. It is estimated based on linear regression model.

Linear regression model:

$$y \sim 1 + x1 + x2$$

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	3100.9	4536.4	0.68355	0.49557
x1	54.775	95.841	0.57152	0.56871
x2	0.41225	0.086491	4.7664	5.2732e-06

Number of observations: 124, Error degrees of freedom: 121

Root Mean Squared Error: 703

R-squared: 0.194, Adjusted R-Squared: 0.18

F-statistic vs. constant model: 14.5, p-value = 2.2e-06

Predicted eCO2: 4909.0021

Alert: Soil moisture is critically low!

eCO2: 400 ppm, TVOC: 0 ppb, Soil Moisture: -47.9605%

Correlation Matrix:

1.0000	0.2081	0.3739
0.2081	1.0000	0.4386
0.3739	0.4386	1.0000

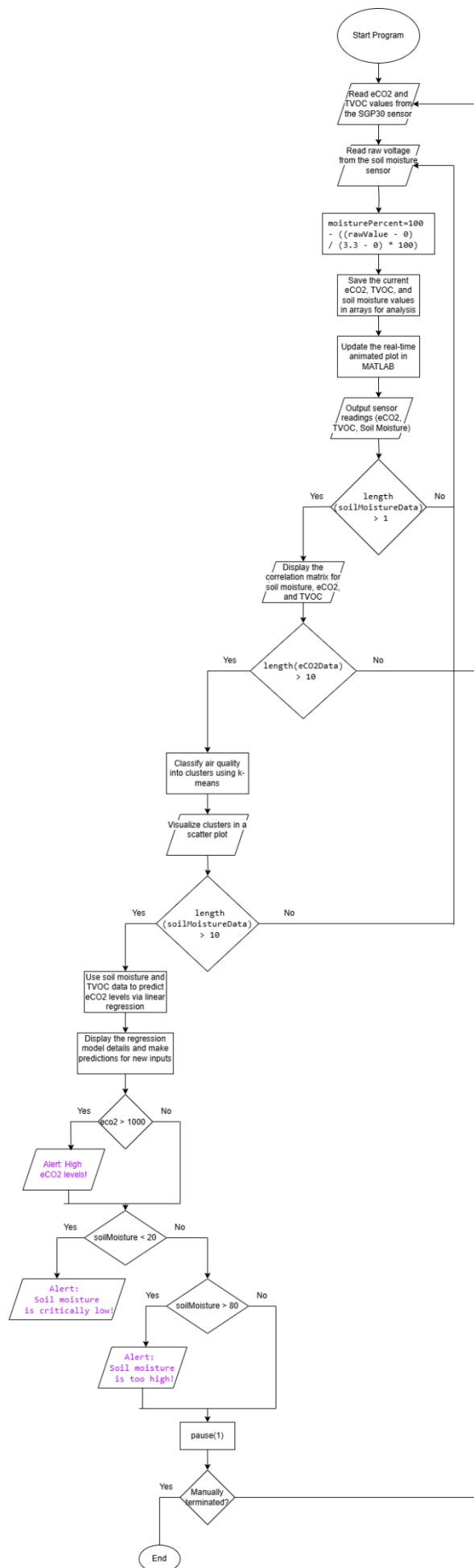


Figure 1. Flow-chart

Contingency

During the execution of this project, several aspects didn't go exactly as initially planned, leading me to adapt and modify my approach to meet the objectives. I expected to gather real-time data from CO₂, PM_{2.5}, PM₁, PM₁₀, CO, O₃ sensors seamlessly and integrate them quickly into MATLAB for continuous monitoring and analysis. However, due to financial and time constraints, I wasn't able to acquire them. That is why I had to use sensors provided in the lab kit (eCO₂, TVOC) that are cheaper, lack the total precision, but have accuracy of 15% within measured value. Next time I will try to find ways to acquire the necessary hardware in advance by consulting professors, TA-s, and recent engineering graduates from York who might have the necessary tools that I need.

Additional material

This project goes beyond simple real-time monitoring by incorporating predictive analysis of eCO₂ levels based on environmental variables like soil moisture and TVOC levels. Predictive modeling is particularly valuable in scenarios where early detection of environmental changes can mitigate risks, such as in agriculture or indoor air quality management. Using MATLAB's linear regression tools, historical sensor data informs a model that forecasts future eCO₂ levels, providing proactive insights. This aligns with studies highlighting the efficacy of predictive analytics in air quality monitoring and environmental forecasting, where machine learning models like regression and neural networks are used to predict air pollutant levels based on correlated factors (Frontiers, 2024).

By anticipating high eCO₂ levels, the system can warn individuals to improve ventilation, reducing exposure to poor air quality. Predictive analytics contributes to smarter urban planning and health interventions (PLOS, 2019).

The integration of predictive tools supports SDG 13: Climate Action by providing actionable data for pollution reduction and mitigation efforts.

Predictive insights can help policymakers enact preemptive measures to address environmental risks, reducing the economic and social costs of reactive interventions (IEEE, 2024).

Conclusion

This project demonstrates the integration of real-time monitoring and predictive analysis as a comprehensive solution for air quality and soil health management. The predictive modeling component enhances the system by providing anticipatory insights, bridging the gap between current conditions and future trends. This innovation not only aligns with sustainable development goals but also underscores the potential for technology-driven solutions to address environmental challenges. Future developments could focus on refining the predictive model with advanced algorithms, such as machine learning, for greater accuracy and broader applications.

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